



SpeakWise: Capturing Real-Time Reasoning Evidence Traces through AI-Mediated Oral Assessment

Jewoong Moon (The University of Alabama) · Yong Ju Jung (The University of Oklahoma)

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Poster Presentation · Work-in-Progress

1 The Problem

Generative AI blurs the line between a learner's own reasoning and AI-assisted production, threatening **assessment validity** for open-ended tasks.

Oral assessment forces real-time articulation (Fenton, 2025), but conventional oral exams **do not scale** and do not capture reasoning as **analyzable evidence**.

2 Approach

SpeakWise is a research-informed intervention grounded in **Evidence-Centered Design** (Mislevy et al., 2003) and assessment design & validation (NRC, 2014), reframing oral assessment as a **process-oriented measurement environment** (cf. Zapata-Rivera et al., 2024).

Evidence-Centered Design

Process data

Human-centred LA

3 Research Questions

- RQ1** What reasoning evidence traces emerge, and how do they vary across learners and tasks?
- RQ2** How do dialogic & temporal patterns shape instructors' interpretation of reasoning?
- RQ3** How do instructors use evidence traces & AI confidence in evaluative judgments?

4 Method

Work-in-progress **design-based research**, synthesizing:

- interaction-log analysis
- automated discourse-pattern detection
- instructor sensemaking interviews

Real-time capture across multiple disciplinary domains.

5 The SpeakWise System

Learners engage in **adaptive oral dialogues** with an AI examiner (Socratic questioning + **tiered scaffolding**). The platform is structured as a four-stage, ECD-aligned pipeline:

1 Oral Dialogue

Student $\rightleftharpoons$ AI examiner · voice · scaffolding

2 Evidence Capture

discourse · dialogic · temporal · structure

3 Reasoning Analysis

4-dim rubric + argument graph + AI confidence

4 Instructor Sensemaking

dashboard review · evidence-linked override

Fig 1. SpeakWise system architecture — an Evidence-Centered Design pipeline from oral dialogue to instructor decision.

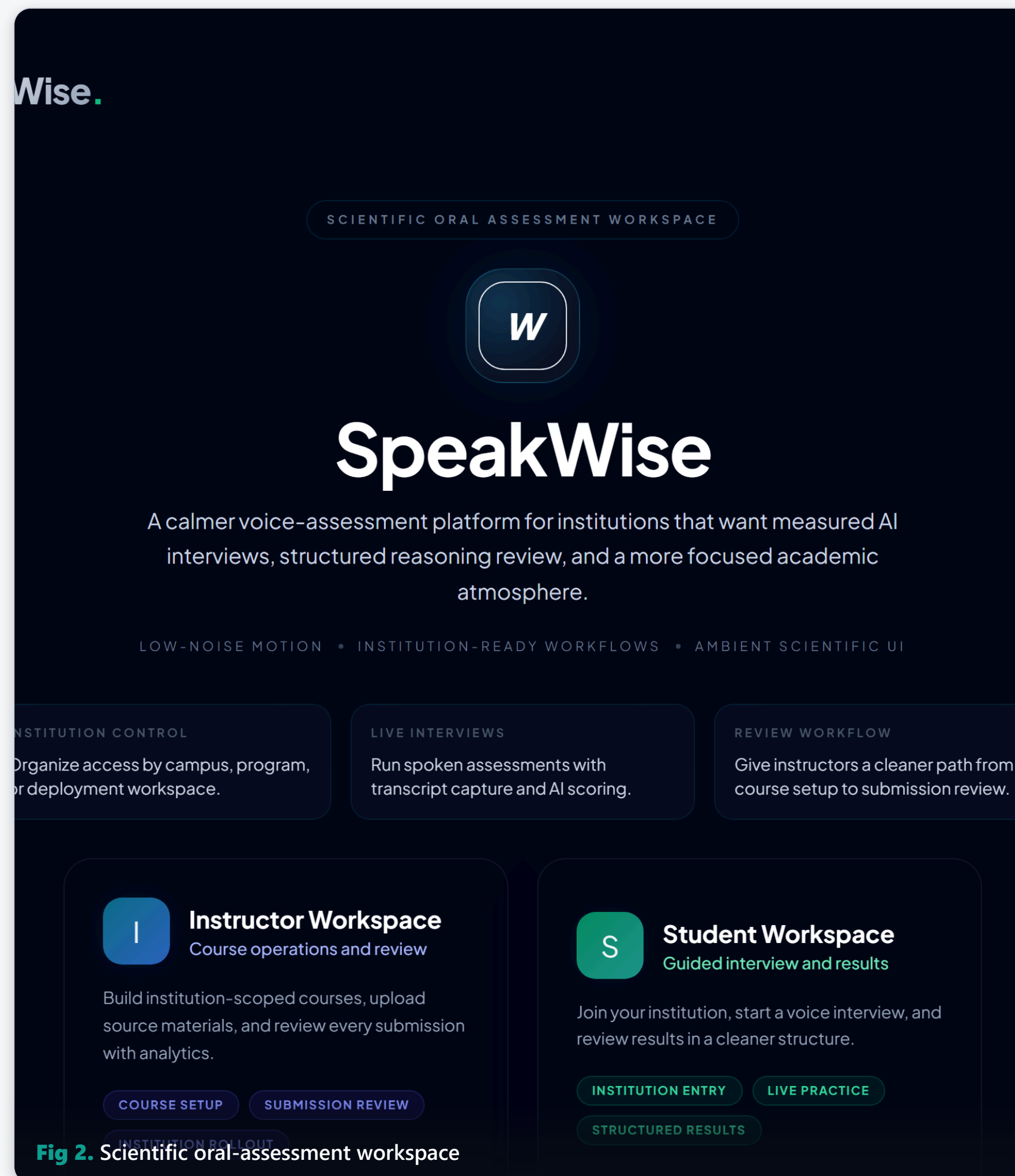


Fig 2. Scientific oral-assessment workspace

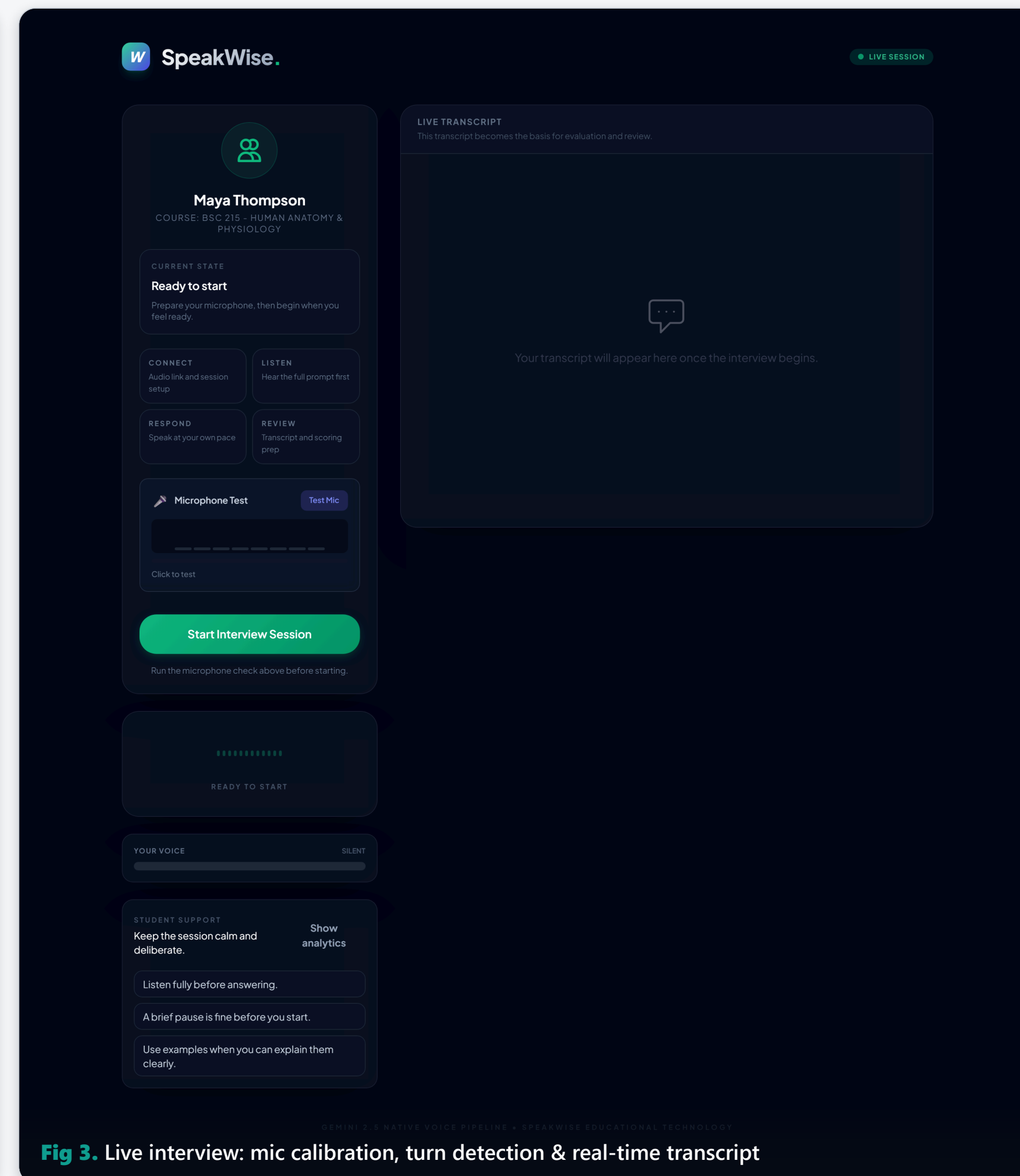


Fig 3. Live interview: mic calibration, turn detection & real-time transcript

6 Reasoning Evidence Traces

The core construct: SpeakWise surfaces **multiple forms of reasoning-relevant evidence** embedded in the interaction (He & Cui, 2025).

- a Justification & causal markers** ("because", "therefore")
- b Counter-argument & abstraction** — challenges, generalizing
- c Dialogic behaviors** — turn initiation, rephrasing
- d Temporal markers** — response-latency variation

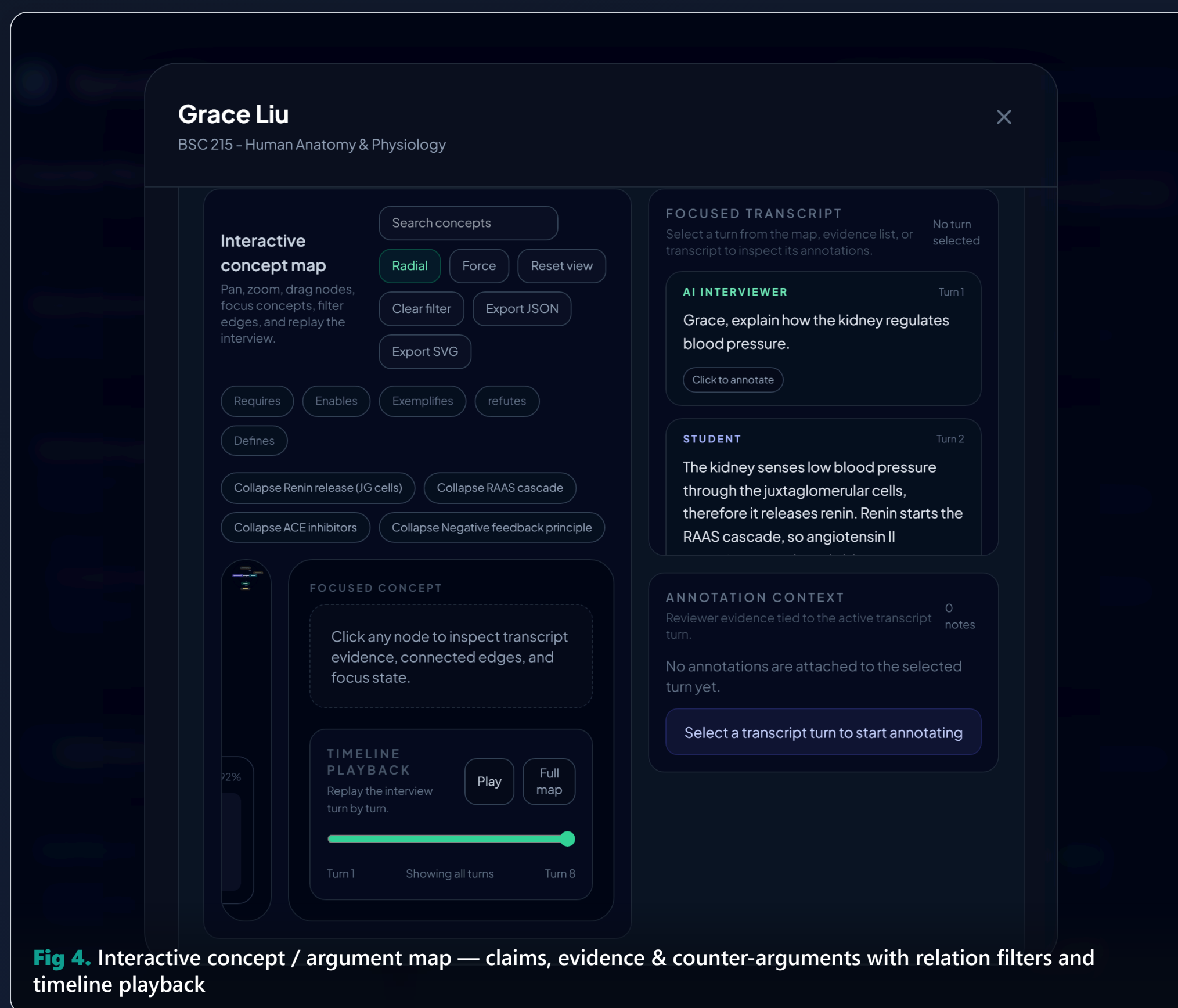


Fig 4. Interactive concept / argument map — claims, evidence & counter-arguments with relation filters and timeline playback

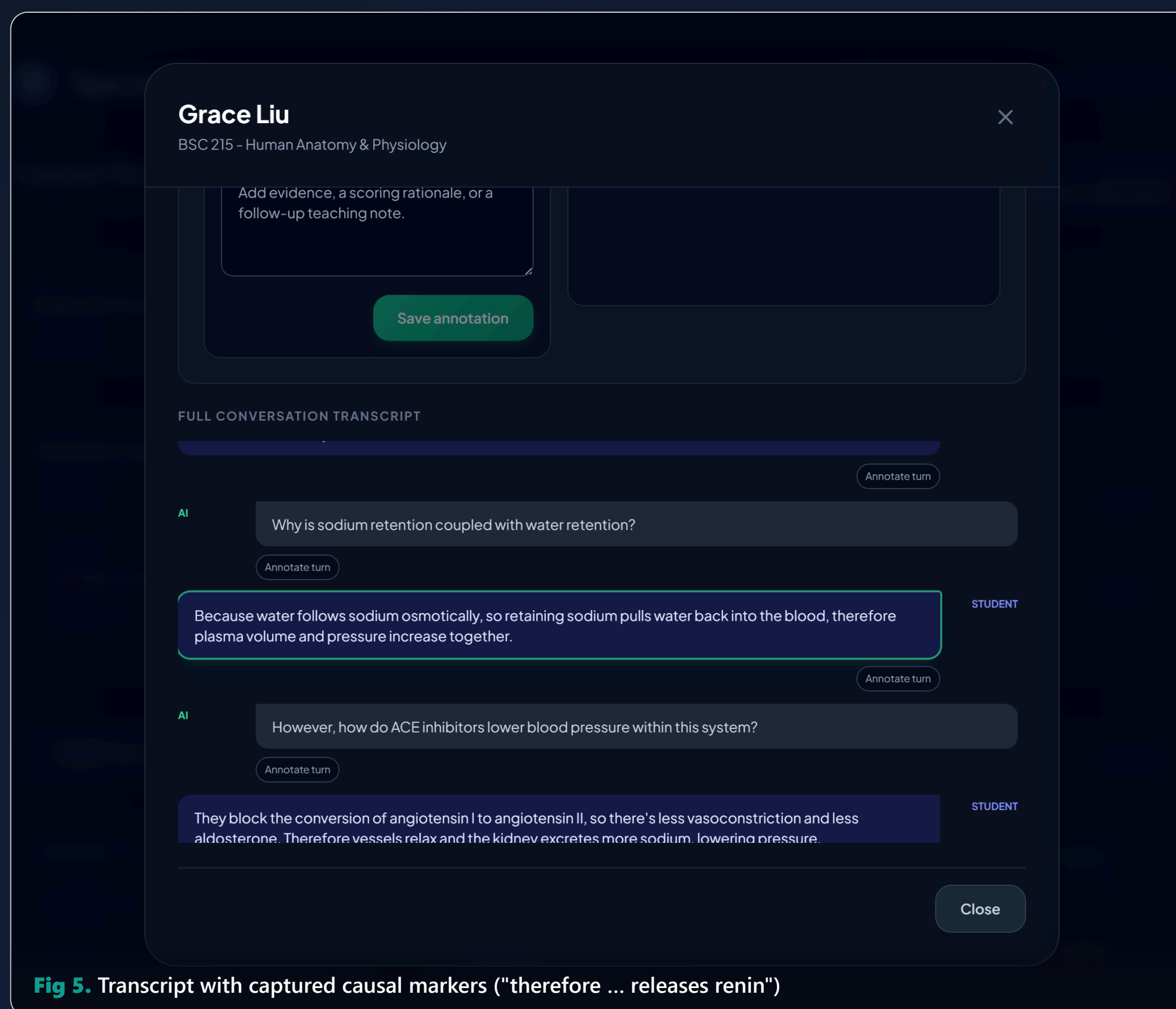


Fig 5. Transcript with captured causal markers ("therefore ... releases renin")

7 Instructor Sensemaking

Automated scores are **provisional summaries, not verdicts** (Alfredo et al., 2024). The instructor surface preserves **agency** by showing not only the score, but how it was generated:

- Confidence-coded feed** — green / yellow / red mastery across submissions
- Attention flags** — courses & students needing first review
- Evidence drill-down** — argument graph + 4-dimension reasoning rubric per student
- Transcript-linked annotations** tied to specific reasoning turns
- Score override** with instructor notes & audit trail

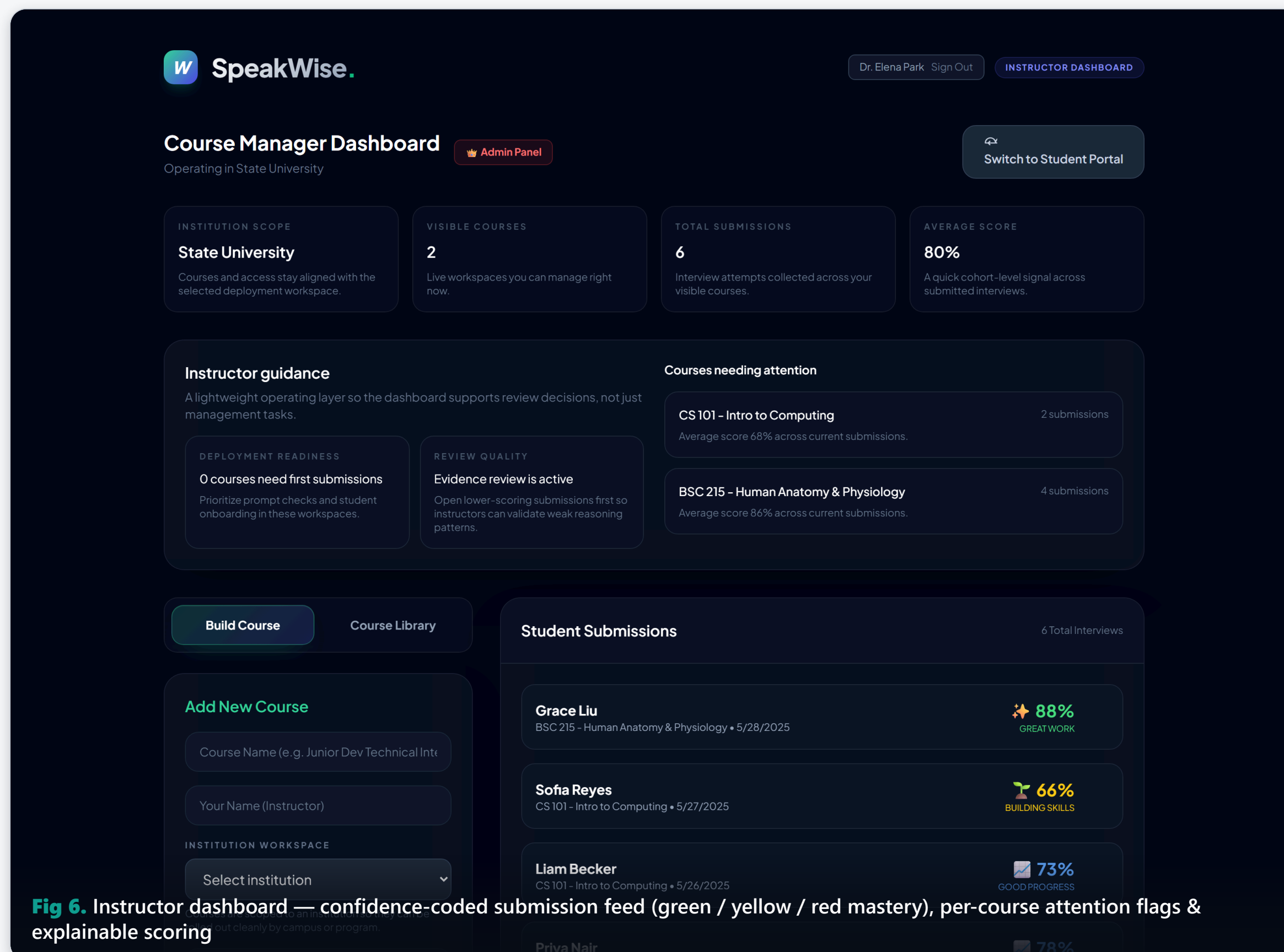


Fig 6. Instructor dashboard — confidence-coded submission feed (green / yellow / red mastery), per-course attention flags & explainable scoring

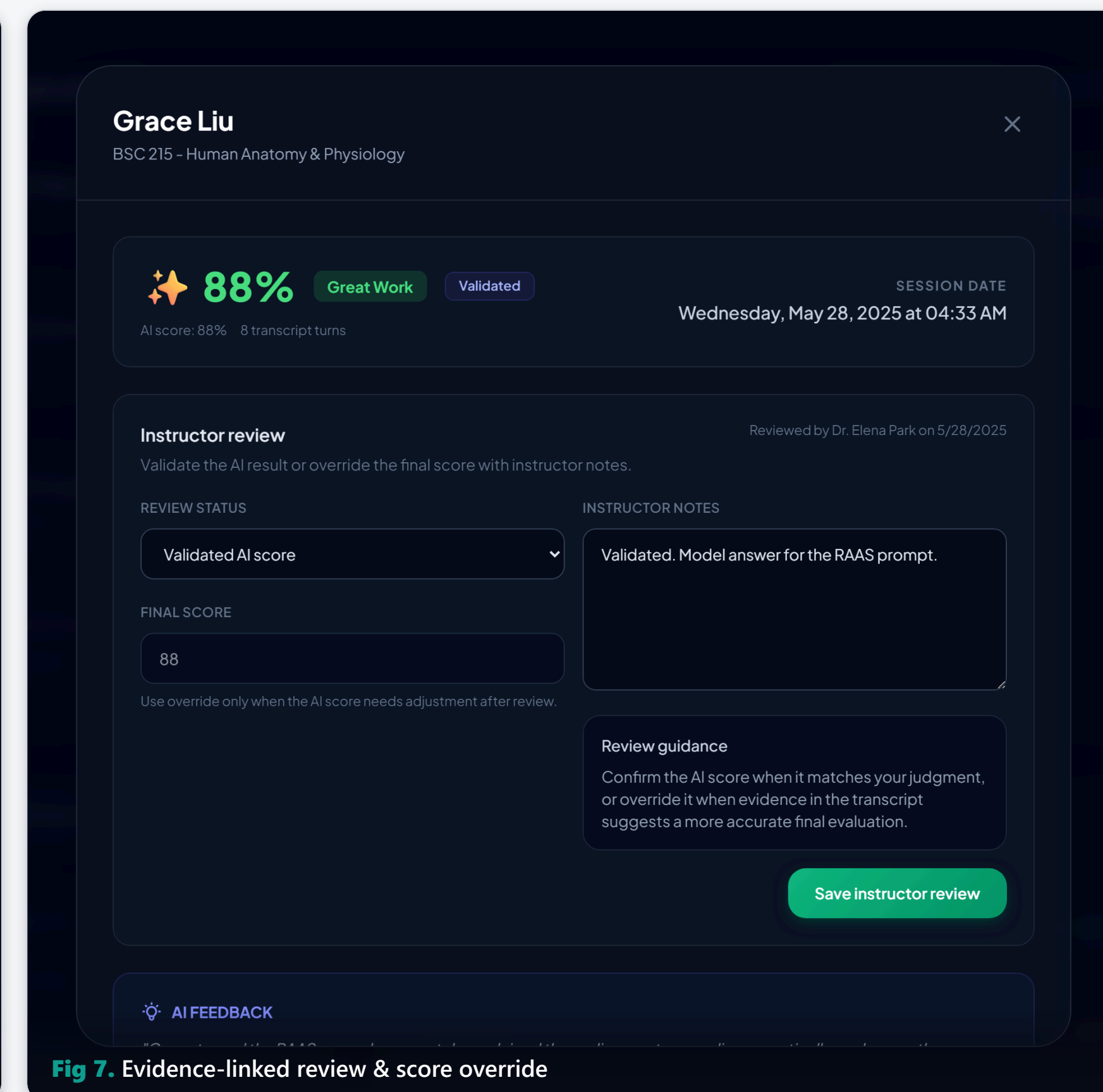


Fig 7. Evidence-linked review & score override

8 Contribution & Next Steps

- Concept** Reasoning evidence traces as a new measurement target for oral assessment, beyond a single holistic score.
- Method** Real-time capture & fusion of **discourse, dialogic, and temporal** signals into an analyzable argument structure.
- Practice** An **institution-ready, explainable, human-in-the-loop** assessment environment that keeps instructors in control.

Next steps: in-the-wild deployment (UA / OU courses) · inter-rater reliability of the reasoning rubric · AI-confidence calibration against instructor judgments.

References

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